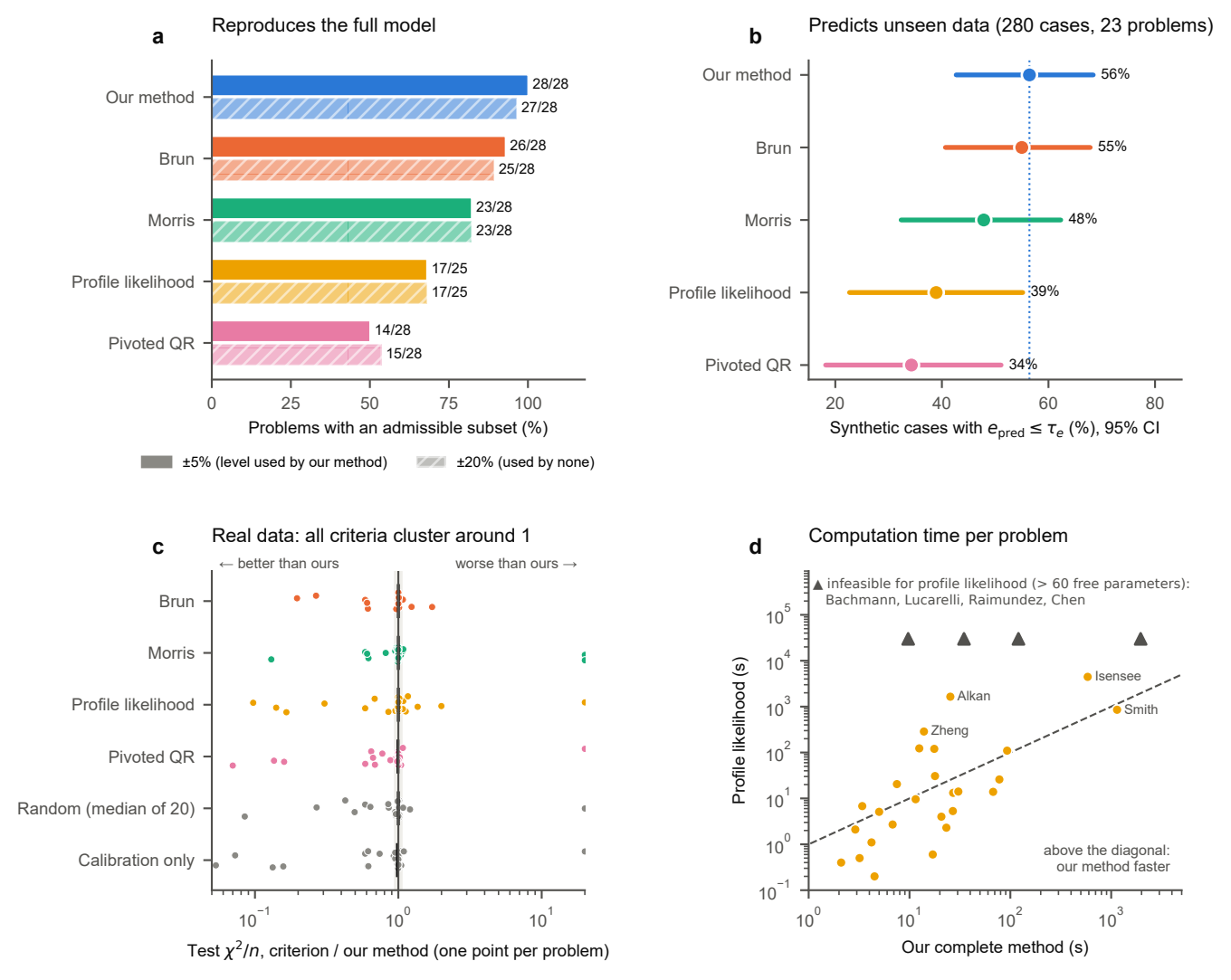

a
Reproduces the full model
Our method
Brun
Morris
Profile likelihood
Pivoted QR
28/28
27/28
26/28
25/28
23/28
23/28
17/25
17/25
14/28
15/28
0
25
50
75
100
Problems with an admissible subset (%)
±5% (level used by our method)
±20% (used by none)
b
Predicts unseen data (280 cases, 23 problems)
Our method
Brun
Morris
Profile likelihood
Pivoted QR
56%
55%
48%
39%
34%
20
40
60
80
Synthetic cases with $e_{pred} \leq \tau_e$ (%), 95% CI
c
Real data: all criteria cluster around 1
← better than ours
worse than ours →
Brun
Morris
Profile likelihood
Pivoted QR
Random (median of 20)
Calibration only
10^{-1}
10^{0}
10^{1}
Test χ^2/n, criterion / our method (one point per problem)
d
Computation time per problem
▲ infeasible for profile likelihood (> 60 free parameters):
Bachmann, Lucarelli, Raimundez, Chen
Isensee
Alkan
Smith
Zheng
above the diagonal:
our method faster
Profile likelihood (s)
Our complete method (s)
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}